Mathematics - K-12 Level Assessment 6 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Restate the expression 4(3 + 2) using the Distributive Property. Do not simplify.
   (a) 4(3) + 2
   (b) 4(3) + 4(2)
   (c) 20
   (d) 3 + 4(2)
2. Which statementis true?
   (a) The product of 5 x 2 is even because both of the factors are even.
   (b) The product of 4 x 4 is odd because both of the factors are even.
   (c) The productof 2 x 7 is even because both of the factors are odd.
   (d) The product of 5 x 3 is odd because both of the factors are odd.
3. Identify the algebraic expression that matches the word phrase: 4 times as many cans as Tom collected.
   (a) 4 t
   (b) t over 4
   (c) t - 4
   (d) t + 4
4. How many positive and negative integers is 12 a multiple of?
   (a) 1
   (b) 4
   (c) 12
   (d) 3
5. Find 1 over 6 + 1 over 8.
   (a) 7 over 24
   (b) 2 over 14
   (c) 1 over 7
   (d) 1 over 4

True / False (10 marks)

Answer True or False

6. True or False: Mark has a total of 73 stamps in his collection from Japan, Canada, and Mexico combined.
7. True or False: The simplified form of the fraction is $\frac{\sqrt{3}}{12}$.
8. True or False: The correct answer to 'Find b if $\log_b 343 = -\frac{3}{2}$.' is '$\frac{1}{49}$'.
9. True or False: Evaluating $(c^2$ - $b^2)^2 + a^2$ for a = 5, b = 3, and c = 2 results in 50.
10. True or False: The mean of the data set 18, 9, 9, 10, 11, 14, 30, 19 is 11.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Stephanie flew 2,448 miles from Los Angeles to New York City. Discuss how you would round this distance to the nearest thousand miles and explain your reasoning.
12. In parallelogram $ABCD$, angle B measures $110°$. Explain how you can determine the measure of angle C and discuss the properties of parallelograms that support your reasoning.

End of Paper